DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

ABES Engineering College, Ghaziabad

B.TECH PROJECT PRESENTATION

CineMatch

Personalized Movie Recommendation System

Academic demonstration of user-based collaborative filtering

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PROGRAM
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Project demonstration • Data, recommendation logic and interactive UI

Project Overview

CineMatch demonstrates how movie ratings can be used to rank unseen movies for a selected demo user.

Purpose

Help users discover movies that may match their preferences by learning from rating patterns.

Core approach

User-based collaborative filtering with cosine similarity, as described in the project deck.

Interface

A Streamlit web application for exploring demo users, rating history and recommendations.

Scope note: the bundled dataset is synthetic and intended for a repeatable offline academic demo—not a real-world benchmark.

Why a Movie Recommendation System?

Large movie catalogues make it difficult to choose what to watch.

- Users face too many movie choices.
- Generic popular lists may not reflect an individual user's taste.
- Past ratings can reveal preference patterns shared by similar users.
- CineMatch ranks unseen movies using these patterns and presents top recommendations.

Project goal: estimate preference scores and recommend top-N unseen movies.

Project Objectives

The prototype demonstrates the core workflow from ratings data to ranked recommendations.

01 **Explore**

Inspect a movie catalogue and a demo user's rating history.

02 **Compare**

Calculate cosine similarity between user rating patterns.

03 **Estimate**

Estimate preference scores for movies the selected user has not rated.

04 **Present**

Show recommendation results in an interactive Streamlit interface.

Previous Work & Related Approaches

Common recommendation methods use user ratings, item attributes, or both.

User-based collaborative filtering

Finds users with similar rating patterns and uses their ratings to estimate a target user's interest in unseen movies.

CineMatch approach

Content-based filtering

Recommends items with similar attributes such as genres, keywords, cast or descriptions.

Needs item metadata

Hybrid recommendation

Combines collaborative signals and item features to balance the limitations of either method alone.

Possible extension

CineMatch focuses on user-based collaborative filtering using cosine similarity; the current demo uses synthetic ratings.

Tools & Technologies

The following stack is listed in the project specification provided for this presentation.

Technology	Role in the project
Python	Core programming language
Pandas and NumPy	Data handling and numerical operations
scikit-learn	Cosine similarity calculation
Streamlit	Interactive web application
CSV files	Lightweight demo data storage

How Recommendations Are Generated

The process described in the project presentation.

1	**Load data**	Movie catalogue + user ratings
2	**Build matrix**	Users × movies, with observed ratings
3	**Measure similarity**	Cosine similarity between user vectors
4	**Estimate scores**	Use similar users' ratings for unseen movies
5	**Rank results**	Return top-N candidate movies

Cosine Similarity

Cosine similarity compares the direction of two rating vectors.

Formula

$$\cos(A, B) = (A \cdot B) / (||A|| \times ||B||)$$

- Each user is represented by a vector of movie ratings.
- More aligned rating vectors indicate more similar patterns.
- Similarity helps estimate preference for movies not yet rated.

This is a similarity measure—not a guarantee that a recommendation will be liked.

Dataset and Demonstration

The supplied project presentation states that the bundled data is synthetic and designed for an offline demo.

Demo users

Select a demo user and inspect their existing rating history.

Synthetic ratings

Ratings are illustrative sample data, not verified real user feedback.

Recommendation output

Generate ranked unseen movies and review estimated scores.

Do not describe the bundled demo ratings as downloaded MovieLens data unless the code/data files confirm that.

Steps & Commands to Start the App

Use the terminal in the folder that contains the application's app.py and requirements.txt.

1	**Open project**	Open the CineMatch project folder in VS Code.
2	**Open terminal**	Choose Terminal → New Terminal.
3	**Activate environment (if .venv exists)**	`.\.venv\Scripts\Activate.ps1`
4	**Install dependencies (first run only)**	`python -m pip install -r requirements.txt`
5	**Run Streamlit app**	`python -m streamlit run app.py`

Open the URL printed by Streamlit (usually http://localhost:8501). Stop the app with Ctrl + C.

Limitations & Future Scope

The current version is a prototype for academic demonstration.

Synthetic data

Results are illustrative and should not be treated as a real-world benchmark.

Cold start & sparsity

New users and sparse rating histories can reduce recommendation quality.

Future improvements

Use a larger real dataset, train/test evaluation, item-based filtering, SVD, accounts and a database.

Conclusion & Questions

CineMatch demonstrates the workflow from movie-rating data to an interactive recommendation interface.

- User-based collaborative filtering ranks unseen movies using rating patterns.
- Cosine similarity measures how aligned two users' rating vectors are.
- Streamlit provides a simple way to demonstrate the workflow in a browser.

Thank you